

Quarterly Plant Inspection Report - Q1 2025

Inspection Date: 15 March 2025 | Inspector: R. Kumar (Senior Mechanical Engineer) | Building B02, North Region

P-101 - Cooling Water Pump [Risk: HIGH]

Vibration at drive end bearing: 6.2 mm/s RMS. Exceeds OISD-137 limit of 4.5 mm/s. Visual inspection reveals slight misalignment of flexible coupling. Last seal replacement on 12 January 2025 per work order WO-20000019. Bearing temperature 71 deg C, within limit. Recommendation: perform laser alignment check within 48 hours. If vibration remains above 4.5 mm/s after alignment, replace drive end bearing 6309-2RS.

P-102 - Feed Transfer Pump [Risk: LOW]

Bearing temperature: 68 deg C. Vibration: 3.1 mm/s, within OISD-137 limit. Seal flush flow confirmed at 2.4 L/min, above minimum 2.0 L/min. No leakage at gland. No issues found. Next inspection: June 2025.

P-201 - Condensate Extraction Pump [Risk: LOW]

Vibration: 2.8 mm/s, well within OISD-137 limit. Bearing temperature 65 deg C. Seal flush pressure 0.95 bar, within 0.8 to 1.2 bar range. No issues found. Next inspection: June 2025.

TK-482 - Feed Storage Tank [Risk: MEDIUM]

Level transmitter LT-482-01 calibration is overdue by 45 days. Last calibration: October 2024. Per Factory Act Schedule VIII, level instruments on process tanks must be calibrated every 6 months. Tank shell shows no visible corrosion. Nitrogen blanket pressure at 0.05 bar. Action required: complete calibration before 30 March 2025.

HX-305 - Shell and Tube Heat Exchanger [Risk: MEDIUM]

Shell side outlet temperature is 12 deg C below design, indicating tube fouling. Pressure drop across tube side increased from 0.8 bar to 1.4 bar since December 2024. Estimated heat transfer efficiency loss: 18 percent. Tube bundle chemical cleaning recommended during next planned shutdown. No tube leakage detected.

BLR-118 - Package Boiler [Risk: CRITICAL]

Pressure test certificate expired on 01 February 2025. Under PESO regulations and Factories Act Section 31, a valid pressure test certificate is mandatory for boiler operation. Operating without a valid certificate constitutes a regulatory violation and exposes the facility to shutdown notice. Immediate action required: contact approved inspection authority. Do not defer beyond 31 March 2025.

R-201 - Process Reactor [Risk: MEDIUM]

Safety relief valve SRV-R201-01 last tested: September 2023. Per OISD-116, safety relief valves on process reactors must be bench-tested every 18 months. Test is due March 2025. Schedule bench test at approved facility before 30 April 2025.

M-017 - Boiler Feed Motor [Risk: LOW]

Insulation resistance: 85 MOhm (healthy, minimum 1 MOhm per IEC 60034). Winding temperature within limits. Terminal connections show no heat discolouration. Vibration at motor bearing: 2.2 mm/s. No issues found.

Inspection Summary

Total assets inspected: 8

Compliant (no issues): 4 - P-102, P-201, M-017, partial HX-305

Action required: 3 - P-101 vibration, TK-482 calibration, R-201 SRV test

Critical immediate action: 1 - BLR-118 pressure certificate

Next full inspection: June 2025